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A Multimodal RAG-Powered System for Visual Trend Detection and Analysis on Social Media

Major Project Phase - II (Review - I)

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Outline

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Introduction to Background

- Social media today is **highly visual**
- Trends often appear through **images before hashtags**
- Current tools analyze mostly text data
- The proposed system detects trends **directly from images** using AI

Motivation and Problem Statement

Motivation

- Social media trends **influence brands, creators, and marketing**
- Detecting trends early gives **higher engagement**

Problem

- Current tools **rely on hashtags and captions**
- Visual trends **appear before text labels**
- Trends are detected **too late**

Objectives

- 1. Detect & Embed** — Identify emerging visual trend clusters from social media images using a CLIP + BERT multimodal pipeline, without relying on textual metadata
- 2. Track & Query** — Monitor temporal cluster evaluation, forecast post engagement trajectories, and enable natural language querying via a RAG module backed by FAISS
- 3. Evaluate** — To evaluate the proposed pipeline on the real time social media data and benchmark the popularity predictor against established baselines.

Gap Identification

Title	Gap Identified
Learning Transferable Visual Models From Natural Language Supervision	CLIP provides strong multimodal embeddings but lacks mechanisms for temporal trend tracking, cluster discovery, and conversational querying over social media data.
BLIP-2: Bootstrapping Language-Image Pre-training with Frozen Image Encoders and Large Language Models	BLIP-2 enables image captioning and visual Q&A but is not designed for unsupervised trend cluster labeling, temporal analysis, or retrieval-augmented querying over social media datasets.
CLIP-Driven Attention Network for Multimodal Sentiment Analysis	Focuses on sentiment classification using CLIP; lacks unsupervised visual trend discovery, temporal tracking, popularity prediction, and RAG-based querying.

Gap Identification

Title	Gap Identified
Improved Baselines with Visual Instruction Tuning	Focuses on visual instruction tuning for general-purpose VQA and conversation tasks; lacks unsupervised trend discovery, temporal cluster tracking, popularity prediction, and RAG-based querying over social media image data.
hdbscan: Hierarchical density based clustering	Provides only a general-purpose clustering algorithm; does not address multimodal visual embeddings, temporal trend tracking, popularity prediction, or RAG-based querying for social media trend analysis.
Asymmetric Discrete Cross-Modal Hashing	Focuses on cross-modal hashing for efficient multimedia retrieval; does not address visual trend discovery, temporal cluster evolution, engagement prediction, or RAG-based querying on social media data.

Gap Identification

Title	Gap Identified
Information Technology for Modelling Social Trends in Telegram Using E5 Vectors and Hybrid Cluster Analysis	Relies solely on text-based embeddings for trend detection, ignoring visual content entirely. TrendLens addresses this by using CLIP to detect visual trends directly from images, capturing aesthetic patterns that emerge before any textual labels or hashtags are assigned.
Large Language Models Are Natural Video Popularity Predictors	Focuses on video-based popularity prediction using LLMs with text-converted frames; does not address unsupervised visual trend cluster discovery from static images, temporal trend emergence detection, or RAG-based natural language querying over discovered trend data — all core to TrendLens.
A Multimodal Approach to Predict Social Media Popularity	Relies on handcrafted features and CNN image scoring for popularity prediction; does not perform unsupervised visual trend discovery, temporal tracking, or RAG-based trend querying.

Gap Identification

Title	Gap Identified
MMSPP: Multimodal Social Media Popularity Prediction	Fuses image, text, and social features to predict retweet/favorite counts for individual posts; does not perform unsupervised visual trend cluster discovery, temporal trend tracking, or RAG-based natural language querying of emerging trends.
Multimodal Deep Learning Framework for Image Popularity Prediction on Social Media	Combines visual and social features via a CNN-based late fusion model to predict per-image popularity scores; does not discover unsupervised visual trend clusters, track temporal trend evolution, or support natural language querying of trends via RAG.
Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks	Proposes RAG for general NLP tasks like open-domain QA using a Wikipedia text index; does not apply retrieval-augmented generation to multimodal visual trend data, unsupervised image cluster discovery, or social media trend querying – the specific domain TrendLens addresses.

Gap Identification

Title	Gap Identified
An Efficient Faiss-Based Search for Mass Spectral Libraries	Optimized for high-speed retrieval and identity matching of chemical spectra but lacks a multimodal reasoning layer to autonomously cluster emerging patterns or interpret unknown data through natural language generation.
Visual Models for Social Media Image Analysis: Groupings, Engagement, Trends, and Rankings	Proposes manual metapicturing layouts for human inspection but requires researchers to manually name and interpret clusters, whereas TrendLens automates this semantic labeling and offers an interactive conversational interface for trend insights.
Instagram - Driven Social Media Trend Forecasting with Learning Algorithm using Live Dataset	Relies on predefined categories and numerical metadata for trend prediction but lacks the ability to autonomously discover hidden visual patterns or provide generative natural language insights into emerging aesthetics.

Mapping Sustainable Development Goals

SDGs	How our project contributes?
SDG 9 – Industry, Innovation and Infrastructure	Our project builds an advanced AI-driven system for analyzing social media trends using multimodal learning and scalable infrastructure.
SDG 12 – Responsible Consumption and Production	By identifying trends early, our system helps industries make smarter, data-backed decisions, reducing wasteful production and overconsumption driven by guesswork.

Proposed Methodology

Stage 1: Multimodal Embedding

- Each image → 512-dim vector via CLIP (ViT-B/32)
- Captions & hashtags encoded in the same semantic space
- Enables cross-modal retrieval and comparison

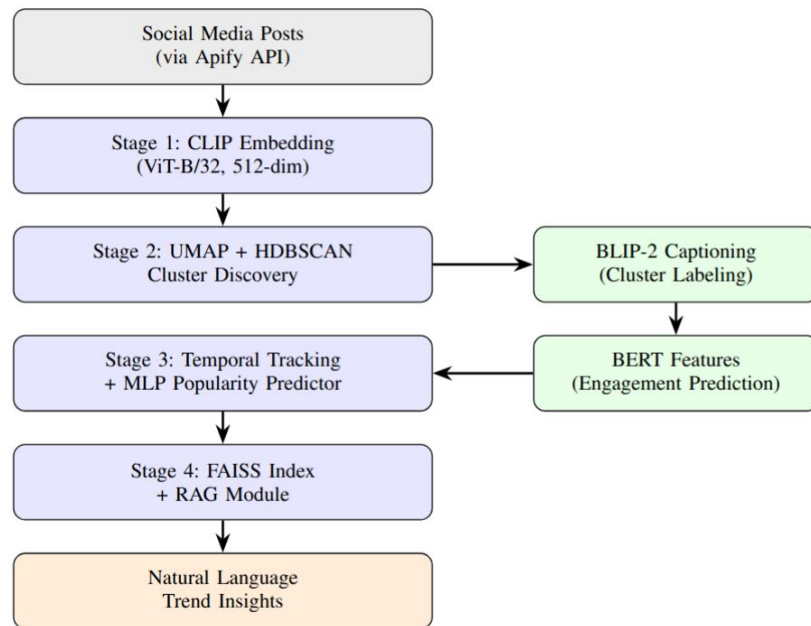


Fig. 1. Four-stage pipeline architecture for visual trend detection.

Proposed Methodology

Stage 2: Cluster Discovery

- UMAP reduces 512-dim vectors to lower dimensions
- HDBSCAN finds arbitrarily shaped clusters — no fixed cluster count needed
- BLIP-2 auto-generates descriptive labels for each cluster

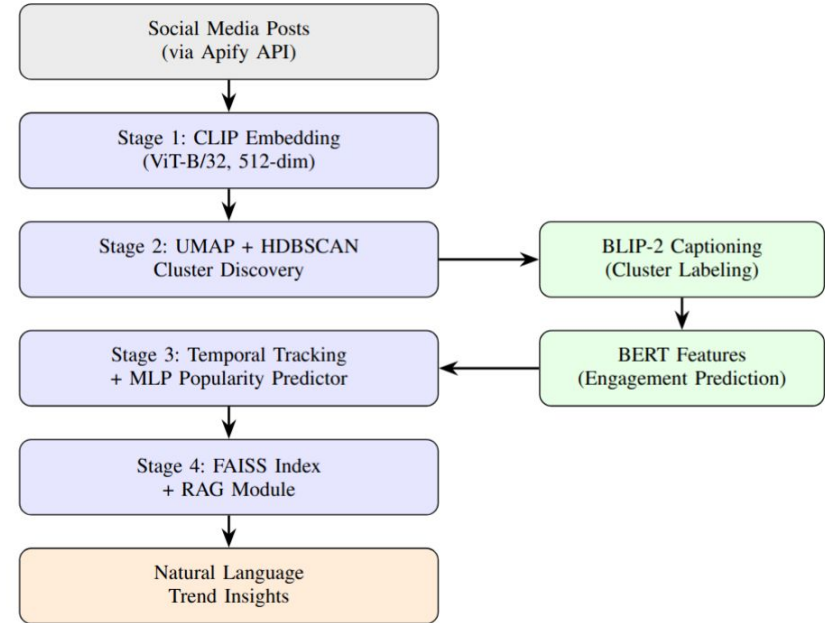


Fig. 1. Four-stage pipeline architecture for visual trend detection.

Proposed Methodology

Stage 3: Temporal Tracking & Popularity Prediction

- Cluster membership indexed against post timestamps → tracks trend growth
- Lightweight MLP trained on CLIP + BERT features
- Predicts engagement score (likes + comments) for new posts

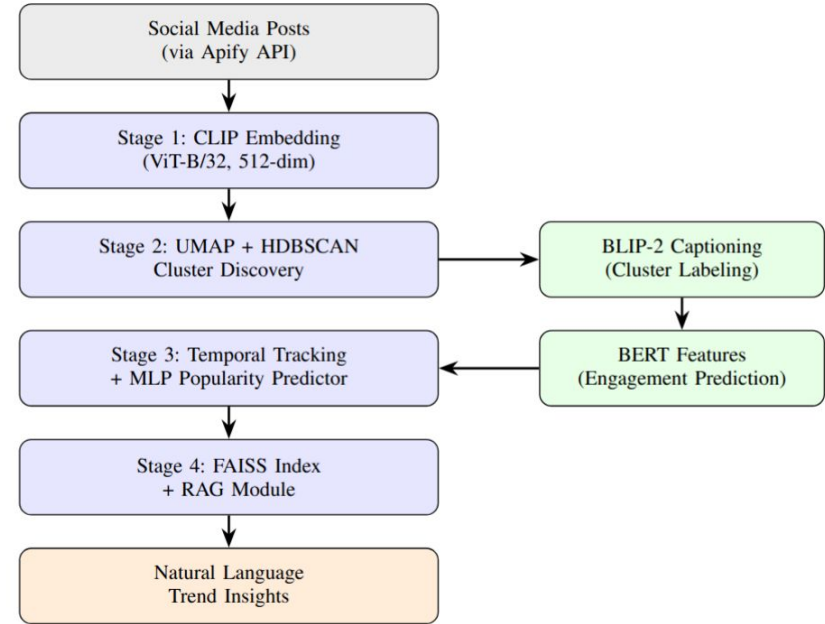


Fig. 1. Four-stage pipeline architecture for visual trend detection.

Proposed Methodology

Stage 4: Retrieval-Augmented Generation

- All cluster captions & metadata indexed in FAISS vector store
- Natural language query → retrieves most relevant clusters
- LLM generates grounded, explainable trend insights

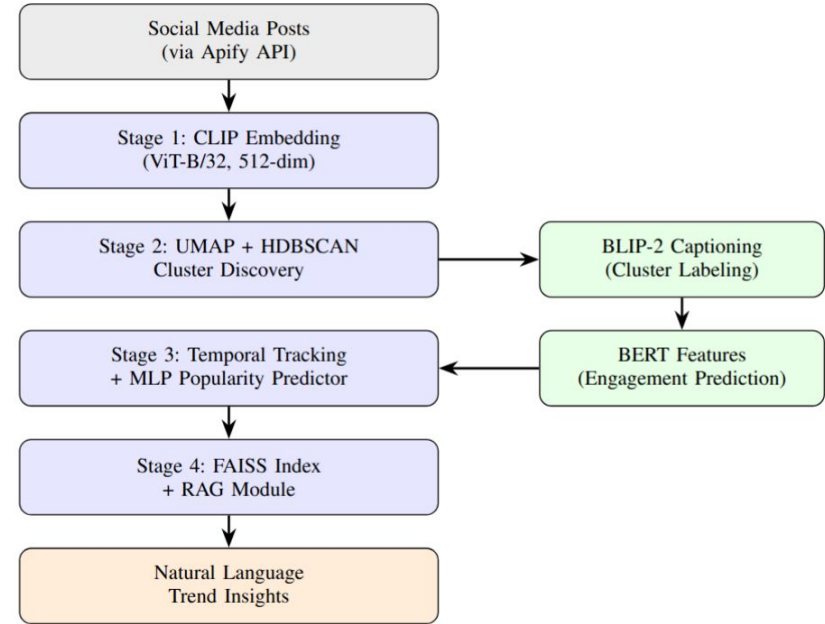
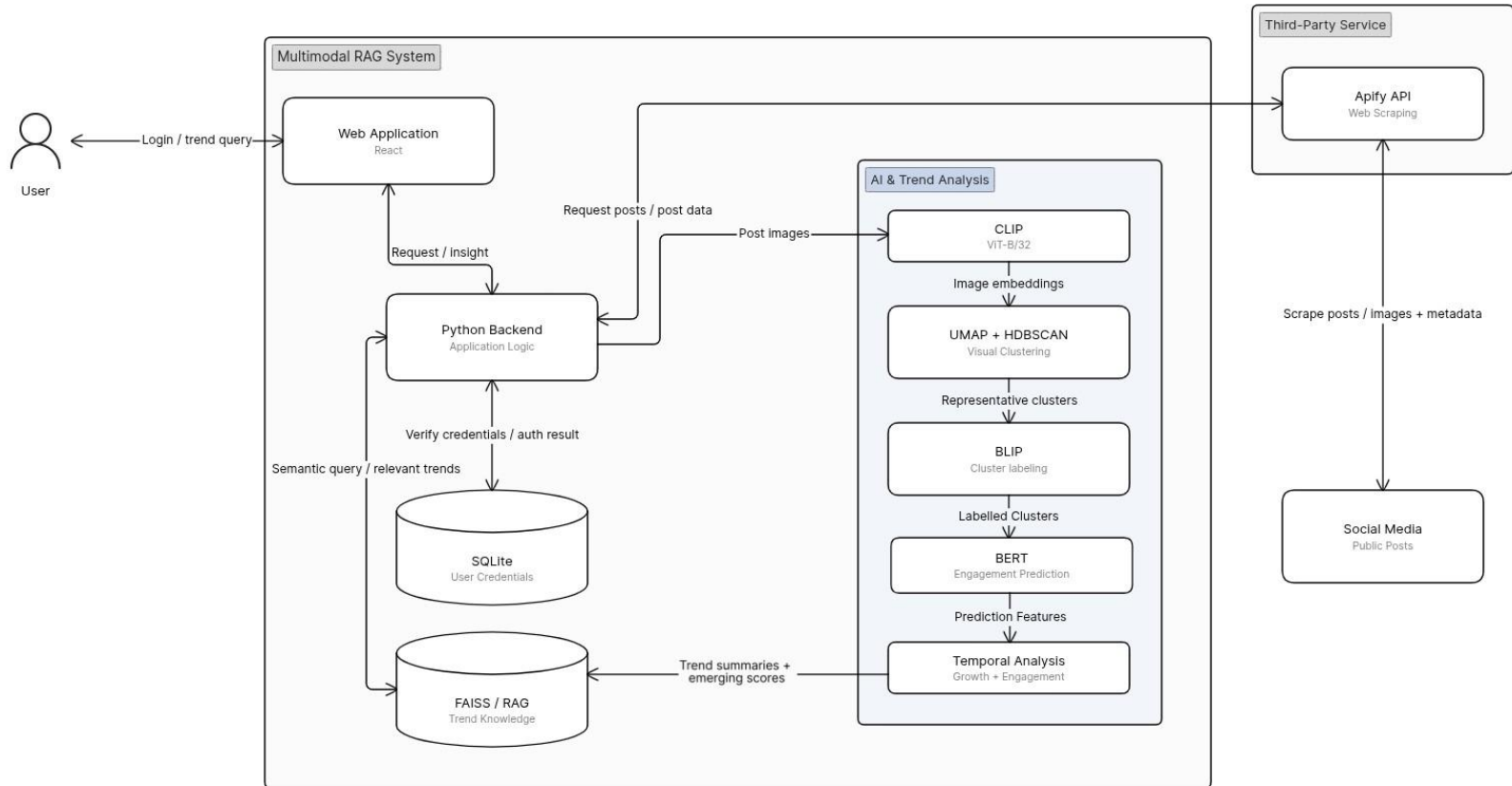


Fig. 1. Four-stage pipeline architecture for visual trend detection.

System Architecture



Results

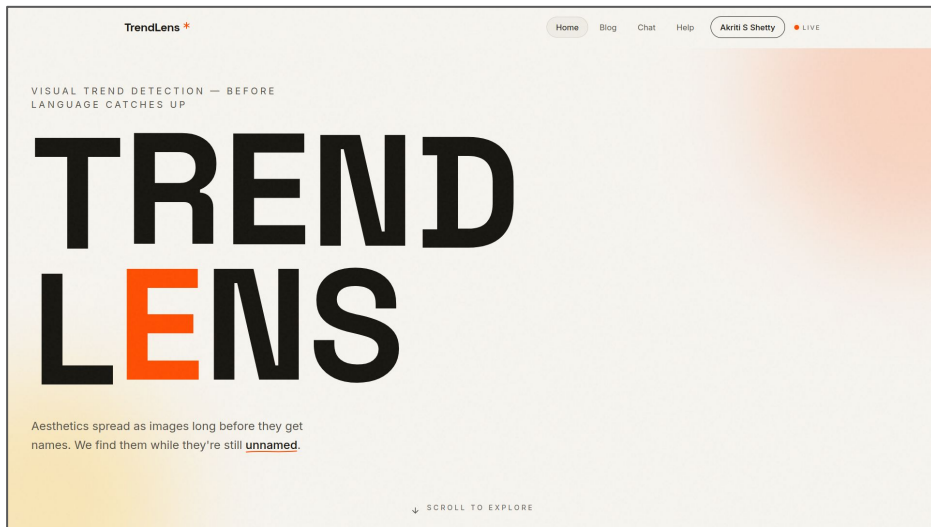


Fig. 3.1. Snapshot of the homepage.

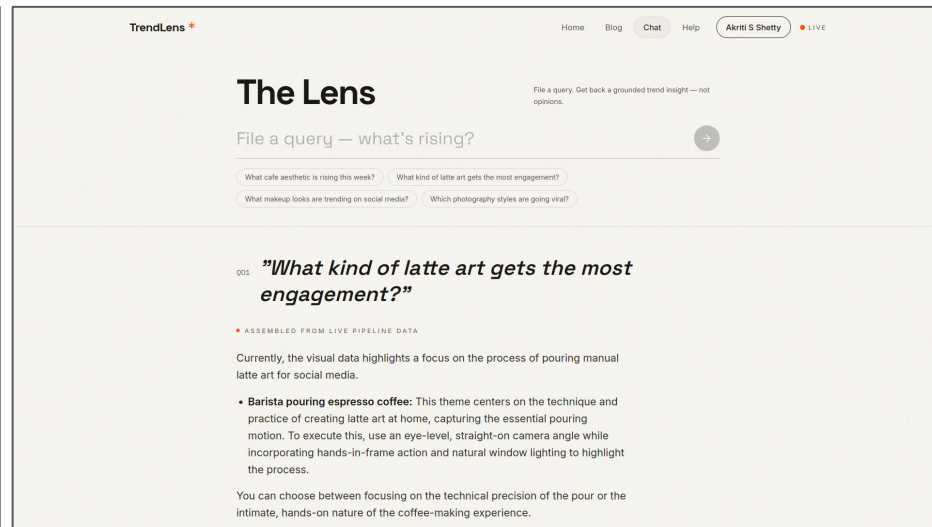


Fig. 3.2. Snapshot of the chat.

Work Pending in Project

- Model and pipeline optimization
- Evaluation against baseline models
- Deployment of the website

Work Pending in Paper

- Result
- Discussion
- Conclusion

References

- [1] Alec Radford., “Learning Transferable Visual Models From Natural Language Supervision” (2021)
- [2] Junnan Li., “BLIP-2: Bootstrapping Language-Image Pre-training with Frozen Image Encoders and Large Language Models” (2023)
- [3] Jialun Lv., “CLIP-Driven Attention Network for Multimodal Sentiment Analysis” (2025)
- [4] Haotian Liu., “Improved Baselines with Visual Instruction Tuning” (2024)
- [5] Leland McInnes., “hdbscan: Hierarchical density based clustering” (2017)

References

- [6] Xin Luo., “Asymmetric Discrete Cross-Modal Hashing” (2018)
- [7] Roman Lynnyk., “Information Technology for Modelling Social Trends in Telegram Using E5 Vectors and Hybrid Cluster Analysis” (2025)
- [8] Pratik Kayal., “Large Language Models Are Natural Video Popularity Predictors” (2025)
- [9] Mayank Meghawat., “A Multimodal Approach to Predict Social Media Popularity” (2018)
- [10] Jianyang Gu., “MMSPP: Multimodal Social Media Popularity Prediction” (2020)

References

- [11] Fatma S. Abousaleh., “Multimodal Deep Learning Framework for Image Popularity Prediction on Social Media” (2021)
- [12] Patrick Lewis., “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks” (2021)
- [13] Chunyuan Qin., “An efficient faiss-based search method for mass spectral library searching.” (2020)
- [14] Gabriele Colombo., “Visual Models for Social Media Image Analysis: Groupings,Engagement, Trends, and Rankings” (2023)
- [15] Nagaraj P., “Instagram - Driven Social Media Trend Forecasting with Learning Algorithm using Live Dataset” (2023)

Thank you

Hyperparameters

Embeddings - CLIP

- `image_resize = (224, 224)`
- `valid_image_exts = {'.jpg', '.jpeg', '.png', '.bmp', '.webp'}`
- `batch_size = 32` (inference)
- `proj_dim = 512`
- Model:
`openai/clip-vit-base-patch32`

Dimensionality Reduction - UMAP

- `n_components = 10`
- `n_neighbors = 15`
- `min_dist = 0.0`
- `metric = "cosine"`
- `seed = 42`

Hyperparameters

Clustering - HDBSCAN

- `min_cluster_size = 10`
- `min_samples = 3`
- `cluster_selection_method = "eom"`
- `k_reps = 9`

Trend Detection (`src/trends.py`)

- `period = "M", window = 3, eps = 1.0`
- `rising_threshold = 0.25,`
`declining_threshold = -0.25`
- Live windows: Instagram
`scan_days=10`

Retrieval - RAG

- `embed_texts batch_size = 16`
- `k = 3 (FAISS top-k), k = 5 (Instagram retrieve)`

Hyperparameters

Captioning — BLIP

- Model:
Salesforce/blip-image-captionin
g-base
- max_new_tokens = 40
- top_k = 6 (keywords per cluster)

Cluster Tracking

- default_assignment_threshold =
0.25 (cosine sim)
- default_micro_cluster_min = 3

Response Generation - LLM

- temperature = 0.7
- maxOutputTokens = 4096

Resources Required

Hardware

- The project requires a computing environment with GPU support for image embedding and captioning tasks.
- Development and experimentation are conducted on Google Colab Free Tier.

Resources Required

Software & Tools

- Python 3.10+ for system development
- PyTorch and Hugging Face Transformers for model loading and inference
- CLIP (Contrastive Language–Image Pretraining) and BLIP-2 for image–text understanding and captioning
- HDBSCAN and UMAP for clustering and dimensionality reduction
- FAISS for vector similarity search
- NumPy and Pandas for data processing and analysis
- Matplotlib and Seaborn for visualization

Resources Required

Dataset

- Uses the Social Media Prediction Dataset (SMPD)
- Contains 486,000+ social media posts
- Data sourced from Flickr
- Includes images, captions, and metadata for social media analysis
- Used for training, clustering, and trend evaluation